

Digital Filters

Physical Measuring Techniques FK7063

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Introduction

In this computational lab, we construct several digital filters using Matlab and compare them to each other. We also analyze data from an EKG signal and determine ways to remove noise from it.

1 Tasks

1.1 Constructing a Second Order Stop-Band Filter

We define the frequency response of a stop-band filter whose gain is 0 at $f = f_0 = 50$ Hz with a bandwidth of $b = 4$ Hz at -3 dB (corresponding to an attenuation of $1/\sqrt{2}$). The sampling rate is $F_s = 1000$ Hz.

In the space of the z-transform, the frequency f_0 corresponds to $\exp(i\Omega_0)$ with

$$\Omega_0 = 2\pi \frac{f_0}{F_s} = \frac{\pi}{10}$$

To get the desired attenuation of 50 Hz, we place a zero at this point, i. e. we add a factor $(1 - \exp(i\Omega_0))$ to the denominator. Since real signals always contain negative frequencies in the same amount as corresponding positive frequencies, we also need to attenuate -50 Hz (corresponding to $-\Omega_0$) in the same way, so we add a factor $(1 - \exp(-i\Omega_0))$.

Now we need to add the property that the filter's maximum gain is 1 and that it has the desired bandwidth. This is achieved by adding a pole (i.e. a factor in the denominator) that is not on the unit circle but still close to the zero that we added. It can be parametrized as $r \exp(i\Omega_0)$ (with r real). For values of z that are far away, the distance to the pole and the zero will be roughly the same, so modulus of the numerator and denominator are equal, leading to a gain of 1. When z approaches the pole and the zero, there comes a point when it is separated from the zero by a distance $d \approx (1 - r)$ and from the pole by a distance $\approx \sqrt{d^2 + (1 - r)^2} \approx \sqrt{2}(1 - r)$. the ratio of the numerator and denominator is then $1/\sqrt{2}$, so the arc around the unit circle where the gain is lower

than -3 dB has a length of $2d \approx 2(1 - r)$. The bandwidth of the filter thus depends on the distance r of the pole to the origin. We require the arc $2d$ to correspond to the bandwidth $b = 4$ Hz:

$$2(1 - r) = 2d \stackrel{!}{=} 2\pi \frac{b}{F_s}$$

$$\implies r = 1 - \frac{b}{F_s} \pi = 1 - \frac{\pi}{250} = 0.98743$$

Again, we add not one but two poles, for positive and negative frequencies. With both zeros and poles, the filter transfer function becomes

$$H(z) = \frac{(1 - \exp(i\Omega_0))(1 - \exp(-i\Omega_0))}{(1 - r \exp(i\Omega_0))(1 - r \exp(-i\Omega_0))}$$

Multiplying out the brackets and then dividing through by z^2 , we get $H(z)$ as a ratio of polynomials in z^{-1} :

$$H(z) = \frac{B(z)}{A(z)} = \frac{1 - 2 \cos(\Omega_0) \cdot z^{-1} + z^{-2}}{1 - 2r \cos(\Omega_0) \cdot z^{-1} + r^2 \cdot z^{-2}}$$

$$= \frac{1 - 1.9021z^{-1} + z^{-2}}{1 - 1.8782z^{-1} + 0.9750z^{-2}}$$

This is the form used by the Matlab function `freqz(b, a, N)` to plot the transfer

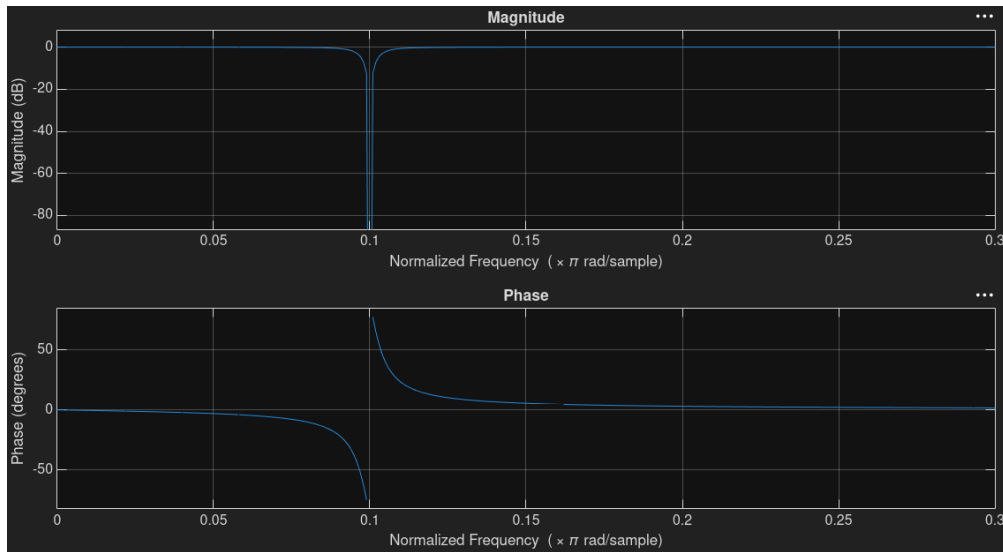


Figure 1: Plot of the transfer function of the second-order stop-band filter, created using the Matlab function `freqz`. The frequency range was chosen to zoom in on the relevant features. We chose 1000 samples for better visibility.

function. b and a specify the coefficients in $B(z)$ and $A(z)$ so $\mathbf{b} = [1 \ -1.9021 \ 1]$ and $\mathbf{a} = [1 \ -1.8782 \ 0.9750]$. This gives the plot shown in Fig. 1. The code is shown in Fig. 2

```

% input parameters
Fs = 1000;
f0 = 50;
b = 4;

% calculate omega0 and r
omega0 = f0/Fs * 2*pi;
r = 1-b/Fs*pi;

% calculate polynomial coefficients
b = [1 -2*cos(omega0) 1];
a = [1 -2*r*cos(omega0) r^2];

% plot using freqz
freqz(b, a, 256);
xlim([0 0.3]);
subplot(2, 1, 2); xlim([0 0.3]);

```

Figure 2: Matlab code to define and plot the second order stop-band filter.

1.2 Second Order Butterworth-Type Stop-Band Filter

Next we construct a stop-band filter with the same characteristics as before, but created as a Butterworth filter using the Matlab function `butter(n, Wn, 'stop')`. `n` is the order (where `n = 1` actually corresponds to a filter where the polynomials are of order 2), and `Wn` is an array containing the break points normalized to the Nyquist frequency. `'stop'` specifies that it is a stop-band filter.

The transfer function plot can be seen in Fig. 3 and the code in Fig. 4.

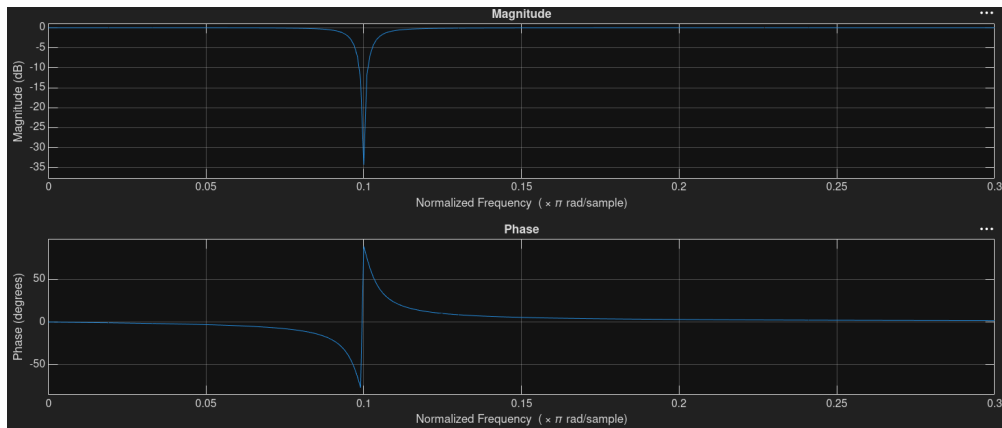


Figure 3: Plot of the transfer function of the second-order butterworth-type stop-band filter.

The transfer functions of the previous filter and the butterworth-type filter have very similar characteristics since they were both defined to have the same stop band with the

```

% input parameters
Fs = 1000;
f0 = 50;
b = 4;

% generate filter
n = 1;
Wn = [f0-b/2 f0+b/2] / (Fs/2);
[b, a] = butter(n, Wn, 'stop');

% plot transfer function
freqz(b, a, 1000)
xlim([0 0.3]);
subplot(2, 1, 2); xlim([0 0.3]);

```

Figure 4: Matlab code to generate and plot a second order butterworth-type stop-band filter.

same bandwidth. Both filters have a jump in phase around the center of the stop-band. The manually defined filter drops more sharply (in theory reaching $-\infty$ dB) in the center of the stop-band because it has a zero there, whereas the butterworth-type filter has a finite minimum.

1.3 Other Filter Types

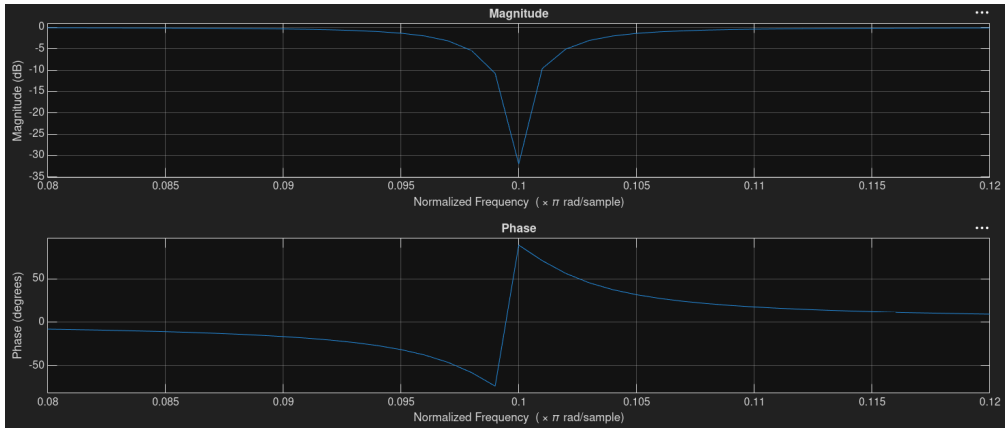
There are several more filter types supported by Matlab. Among them are

- Chebyshev type I: `cheby1(n,Rp,Wn,'stop');`
defined with pass-band ripple R_p in dB.
- Chebyshev type II: `cheby2(n,Rs,Wn,'stop');`
defined with stop-band attenuation R_s in dB.
- Elliptical: `ellip(n,Rp,Rs,Wn,'stop');`
defined with both R_p and R_s .

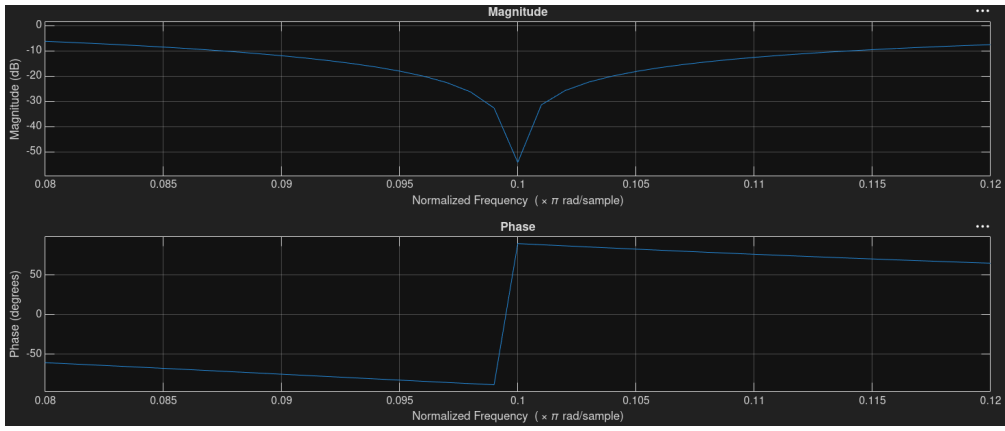
For each of these filters we replace the function `butter(...)` in Fig. 4 with one of the above, choosing $R_p = 2$ and $R_s = 20$. Also, we again choose $n = 1$ which results in second-order polynomials. The transfer functions are shown in Fig. 5.

The Chebyshev type I and elliptical filter are identical (this can be seen by looking at the plots and also by comparing the polynomial coefficients). When changing the R_s value for the elliptical filter, we see that the coefficients always stay the same, so at this order and with $R_p = 2$ the stop-band attenuation actually has no impact on the elliptical filter and it is identical to a Chebyshev type I filter.

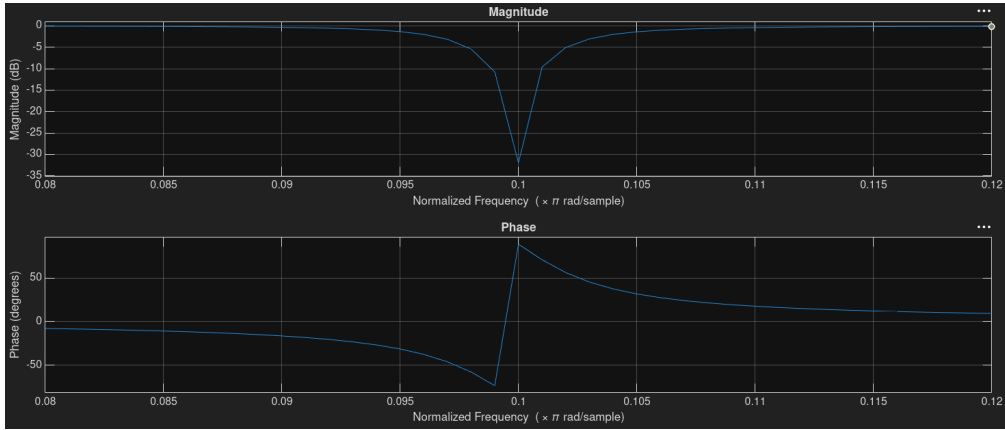
The Chebyshev type II filter reaches a much lower gain (~ -50 dB) than the other two but also has a much slower rise away from the stop band. In fact, it does not adhere to the provided breaking points which likely means that the chosen filter order was too low.



(a) Chebyshev type I filter.



(b) Chebyshev type II filter.



(c) Elliptical filter.

Figure 5: Transfer functions of three different kinds of stop-band filters. The frequency ranges of the plots are chosen close to the stop-band to emphasize differences in the frequency response.

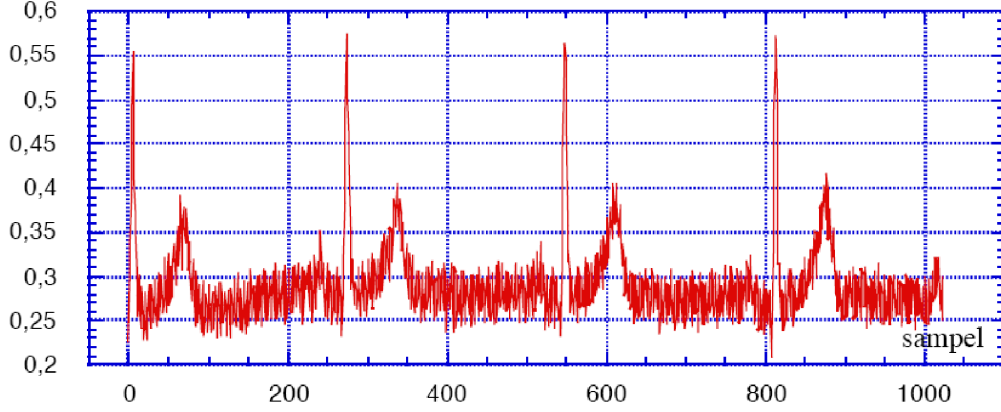


Figure 6: EKG signal with noise [1].

2 Sampling an EKG signal

Next, we consider data from an EKG. The first example has a sampling rate $F_s = 300$ Hz with $N = 1024$ samples, see Fig. 6.

The maximum frequency without aliasing is the Nyquist frequency, i.e. half of the sampling rate:

$$F_{\max} = \frac{F_s}{2} = 150 \text{ Hz}$$

The total sampling time is $T = N/F_s$ and the frequency resolution of the signal's Fourier transform would be $\Delta f = \frac{1}{T} = F_s/N \approx 0.293$ Hz. The frequency with the largest amplitude is likely that of the recurring heartbeat signal. There are about 3 beats between samples 0 and 800, so the frequency is

$$f_H = \frac{3}{800/F_s} = 1.125 \text{ Hz}$$

The data we are actually provided is different from the provided plot in Fig. 7. We will analyze the data from `EKG_02.wav` which contains $N = 82893$ samples at a sampling rate $F_s = 8000$ Hz so the data spans $T = N/F_s = 10.36$ s. It is plotted in Fig. 7.

To analyze the frequency content, we take the signal's fourier transform. The frequency resolution is $\Delta f = 1/T = 0.0965$ Hz and the Nyquist frequency is 4000 Hz. The fourier transform is plotted in Fig. 8.

We can see that the dominant frequency is actually not that of the heartbeat, but the oscillation at 50 Hz. This is likely caused by the mains voltage. We can filter out this unwanted signal by applying the stop band filter from Sec. 1.1. this is done by taking `x = filter(b, a, x)` where `b` and `a` are the arrays given in Sec. 1.1. The filtered signal is shown in Fig. 9. There is now much more structure visible because the interfering 50 Hz oscillation is removed. The most important characteristic of the signal is the sharp drop and then peak in voltage caused by the heartbeat. This is followed about 0.3 s later by a second, much smaller peak.

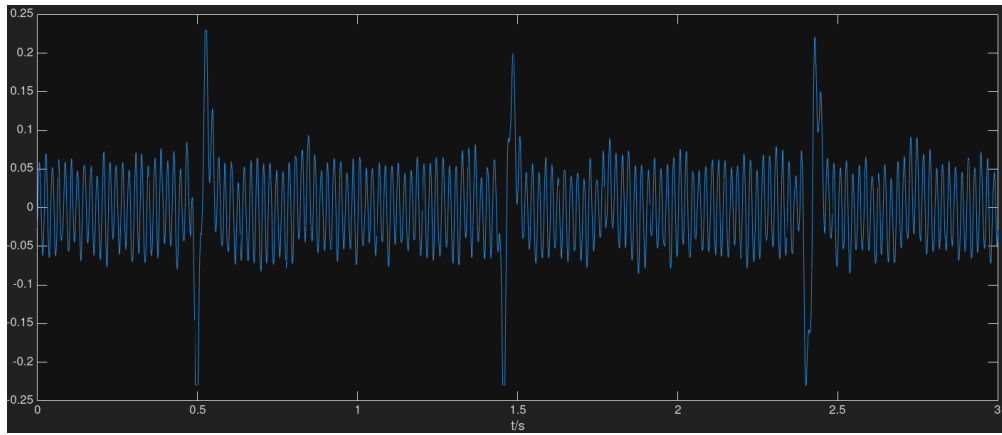


Figure 7: Plot of the data from EKG_02.wav.

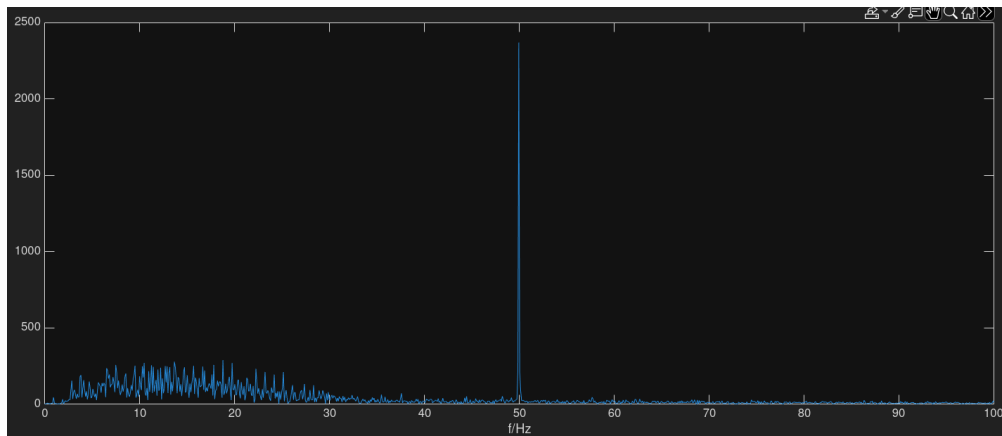


Figure 8: Fourier transform of the data from EKG_02.wav.

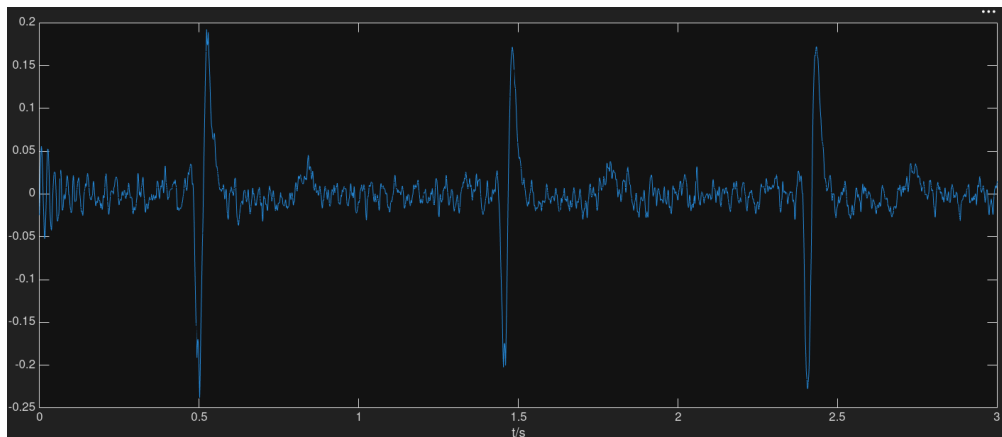


Figure 9: Plot of the data from EKG_02.wav with a 50 Hz spot-band filter applied.

3 Conclusion

In this computational lab, multiple digital filters were constructed using Matlab. The task was to construct stop-band filters with the same stop band characteristics using different methods. We constructed a filter by placing specific poles and zeros. This filter had the best characteristics in terms of attenuation of the stop frequency. Other filter types used were Chebyshev type I and II and an elliptical filter.

Additionally, an EKG signal was analyzed by looking at its Fourier transform. We could see a strong contribution at 50 Hz likely caused by the mains voltage. After filtering this out, we identified two distinct peaks for each heartbeat.

References

- [1] *Digital Filters*. Laboratory exercise manual. 2026.